

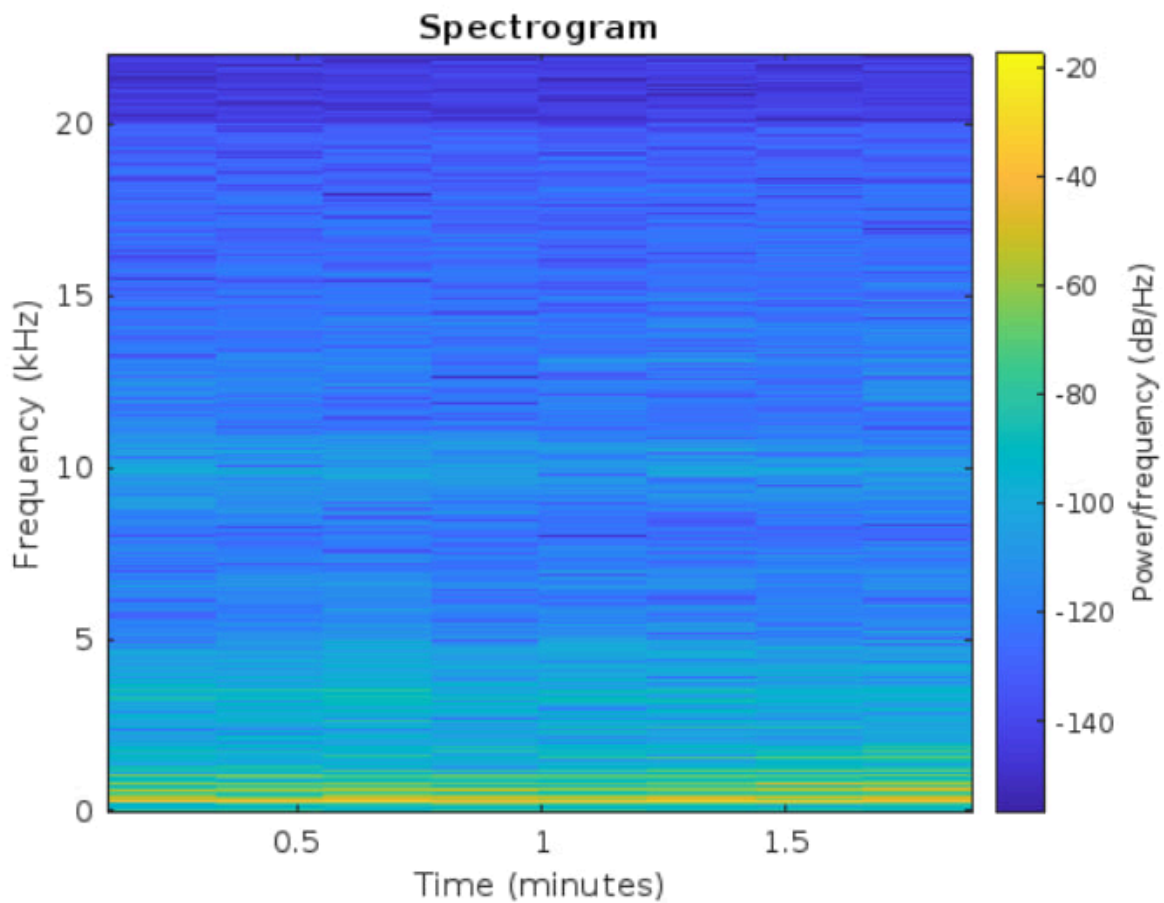
Task 4

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A.

**Can you identify sections where high or low frequency content suddenly changes?
Do these correspond to different instruments or melody changes?**

This is the plot output:



Using MATLAB's `spectrogram()` function on the original music file gives insightful information on the time-dependent frequency content of the signal. The following conclusions can be drawn from the spectrogram:

1. Time Changes in Frequency Energy

The spectrogram reveals clear changes in spectral energy over time. Some time intervals possess:

- An abrupt reduction or increase in low-frequency energy (below 1 kHz), which will generally coincide with changes in basslines or lower-pitched instruments such as cello or piano.
- Sudden increases in high-frequency energy, especially above 5 kHz, corresponding to percussive attacks, harmonic overtones, or sudden melodic transitions.

2. Musical Sections & Instrumentation

Distinct musical sections are depicted in the spectrogram as follows:

- 0 – 0.8 seconds:
We can notice a concentration of energy below 1 kHz, which is likely the entrance of low-end harmonic instruments such as a piano or synthesized bass pad. The absence of high-frequency content suggests a soft or mellow opening section.
- At approximately 0.9 seconds:
We notice a dramatic rise in spectral energy that emanates from approximately 2 kHz to 8 kHz. This is a harmonic or melodic movement, and it could involve strings or ambient textures introducing additional tonal color.
- Following 0.9 seconds:
We notice sporadic bursts of energy along a wide frequency range, which might translate to rhythmic articulation or dynamic entrances and exits of instruments.

3. Noise & Harmonics

Although the original audio itself is fairly clean, in some regions there are transient harmonic peaks higher than 10 kHz. These may be caused by instrumental overtones, artificial reverb tails, or possibly background studio artifacts.

These high-frequency elements do not control the mix but are important for preserving the natural spatial character of the recording.

4. Interpretation of Sound Events

Several obvious musical changes can be related to the spectral data:

Low-frequency buildup: Most likely indicates the entrance or swelling of lengthy instruments.

Mid-to-high-frequency expansions: Represent melodic or harmonic growth, or the entrance of instruments like violin, synths, or percussive elements.

Silence or the absence of spectral activity: May suggest transitions, breaks, or dynamic breakdown in the music.

The spectrogram analysis offers a handy visual observation of how the frequency content evolves in the piece. It highlights:

- Instrumental layering
- Structural sections of music
- The balance of low, mid, and high-frequency content.

These observations are well within normal compositional practices in cinematic or ambient musical genres.